The University of Georgia

ECSE 2920: Design Methodology

User Manual

Skeleton/Outline Version 2

Group 5

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Basics (Features)

Current Deliverable Design

Digital Potentiometer Circuit

For the Digital Potentiometer circuit, the Digital Potentiometer is controlled through a Python script and, through user input. All it does is change the resistance across Terminal A and the Wiper pins. The proper usage would be running the circuit with the Raspberry PI powered on and using a Multimeter to measure the resistance to display that the Digi Pot is working correctly, and the resistance is changing to needed values.

Rotary Encoder Circuit

For the Rotary Encoder Circuit, the physical rotary encoder is pressed, when the PI is on, and the circuit is run, which turns the LED on and off successfully.

Safety Information

Make sure the Raspberry Pi is powered off and/or unplugged before making changes or adjusting the live circuit.

Make sure that a live wire is not touching any metal part of the Pi Hat protector or Raspberry Pi.

Ensure that the device itself is adequately cooled.

List of Components

1. General Parts				
Part	Comments	Quantity	Student Replace	Link
Dupont Wires	FF,FM,MM	120 (40 each)	Return	
Wire Jumper kit		1	Return	
Breadboard kit (Rexqualis)		1	Return	https://www.amazon.com/Breadboards-Solderless-Breadboard-Distribution-Connecting/dp/B07DL13RZH/ref=sr_1_16?crid=2GBGFKZKZRKEM&keywords=proto+board&qid=1682345937&sprefix=proto+board%2Caps%2C105&sr=8-16
Pi Hat Protector	2 fuses	1		
24V Power Supply		1	Yes	https://www.amazon.com/ALITOVE-100-240V-Adapter-Convertor-5-5x2-1mm/dp/B01GC6VS8I/ref=sr_1_3?adgrpid=134470329143772&hvadid=84044627575314&hvbmt=be&hvdevc=c&hvlocphy=45874&hvnetw=o&hvqmt=e&hvtargid=kwd-840443127908:loc-190&hydadcr=24656_10676145&keywords=24v%2B5a%2Bpower%2Bsupply&qid=1704854742&sr=8-3&th=1
Jack Adapter		1	Yes	https://www.amazon.com/Komile%C2%AE-Female-2-1x5-5mm-Adaptor-Connector/dp/B015OCV5Y8/ref=dp_prsubs_3?pd_rd_i=B015OCV5Y8&psc=1
Quad Comparator	LM339	2		
Quad Op Amp	TL074/TL084	1		
Single Op Amp	TL081	2		
NPN Transistor		2		
PCB Standoffs		kit		
10k Potentiometer		1		
IRF4905	P Channel Mosfet	1	Yes	https://www.amazon.com/BOJACK-IRF4905-Transistors-IRF4905-P-Channel/dp/B0854CGZDQ/ref=sr_1_1_sspa?adgrpid=134580290437771&hvadid=84112915260439&hvbmt=be&hvdevc=c&hvlocphy=45874&hvnetw=o&hvqmt=e&hvtargid=kwd-841313956660:loc-190&hydadcr=3016_13539410&keywords=irf4905+mosfet&qid=1704855032&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1
IRFZ34	N Channel Mosfet	1	Yes	https://www.amazon.com/Bridgold-IRFZ34N-IRFZ34-N-Channel-Transistor/dp/B07TVJWXPQ/ref=sr_1_3?crid=1KHPMAUOSS8VT&keywords=irfz34+mosfet&qid=1704855105&sprefix=irfz34+mosfet%2Caps%2C104&sr=8-3
Shunt Resistor		1		Get more as needed
LED Assortment		6		Use labs
Stripboard		Optional		https://www.amazon.com/ELEGOO-Prototype-Soldering-Compatible-Arduino/dp/B0722ZY19F/ref=sr_1_1_sspa?dchild=1&keywords=perfboard&qid=1619107723&sr=8-1-spons&psc=1&spLa=ZW5jcnlwdGVkUXVlbGlmaWVyPUFYVE5JCVTUxNDMzTUcmZW5jcnlwdGVkSWQ9QTAzNzAMMlRTFRGNUNTUESESgmZW5jcnlwdGVkQWRJZD1BMDQzNTMxMzFFWVlTUDJaNDdHTCZ3aWRnZXROYW1lPXNwX2F0ZlhY3Rpb249Y2xpY2tSZWRpcmVjdCZkb05vdExvZ0NsaWNrPXRydWU=
Assortment of Resistors				
Assortment of Capacitors		Kit will be placed in fab lab		

2. Test Bench Parts			
Part	Quantity	Student Replacement	Link
MCP4231 (Dual Digipot)	1	Yes	Find on Digikey
4131-103 (Digipot)	2	Yes	Find on Digikey
Banana Plug Receptacle	4		
20x4 LCD	1	Yes	20x4 Character LCD Display for Arduino	Jameco Valuepro
Rotary Encoder (KY-040)	2	Yes	

3. Canakit Parts			
Part	Quantity	Student Replace	Link
Raspberry Pi 4 (4GB)	1	Yes	https://www.canakit.com/raspberry-pi-4-basic-kit.html
Micro HDMI	1	Yes	https://www.canakit.com/raspberry-4-mico-hdmi-cable.html
Heat sinks	3	Yes	https://www.canakit.com/raspberry-pi-4-heat-sinks.html
USB MicroSD Card Reader	1	Yes	https://www.canakit.com/mini-micro-sd-usb-reader.html
32GB MicroSD Card	1	Yes	https://www.amazon.com/SanDisk-Extreme-microSDHC-UHS-3-SDS
Pi On/Off Switch	1	Yes	https://www.amazon.com/LoyoRPi-MicroUSB-Switch-Raspberry-Fam
Pi Power Supply	1	Yes	https://www.amazon.com/CanaKit-Raspberry-Power-Supply-USB-Cd
GPIO Breakout with ribbon cable	1	Yes	https://www.amazon.com/KEYESTUDIO-GPIO-Breakout-Kit-Raspber
Raspberry Pi Mouse	Optional	Yes	https://www.canakit.com/official-raspberry-pi-keyboard-mouse.html
Raspberry Pi Keyboard	Optional	Yes	https://www.canakit.com/official-raspberry-pi-keyboard-mouse.html

Connecting the Raspberry Pi 4

Circuit control script should be auto set up to be run at boot/plug in.

Using the Electronics Tool Bench

Digital Potentiometer

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User Interface

FAQ

N/A

To do rubric for final design:

Write a complete step-by-step document instructing a novice user how to use your project. Think about
your project as a product bought by the user. The manual will instruct the user on all of the functions of
your design. Hint: Drawing on your own experiences of reading (or deciding not to read user manuals)
will help your team create an interesting piece.

● From your own experience recall that, pictures, drawings, diagrams, etc… make a user's manual
much more beneficial.

● User manuals have several different sections, so don't forget to think about a section on errors or what might go wrong, faqs, clarification, etc…

● User Manuals also contain safety information for the user.

● Your user manual may also include informative Youtube videos or other informative outlets to aid the user.

● Use your creativity…

o Websites

o Videos

o Your other cool ideas….